Name :- Rutuja uday patil
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LAB EXERCISE 1

UNIT 1

1)Title : Study of data types

a) Write a program to display different data types(numbers,strings,lists tuples ,dictionaries)

#number

```
num_int=25
num_float=10.75
num_complex=7+5j
print("numbers")
print("integer :",num_int)
print("float :",num_float)
print("complex :",num_complex)
```

Output :-

```
In [3]: runfile('C:/Rutuja patil/python1/num.py', wdir='C:/Rutuja patil/python1')
numbers
integer : 25
float : 10.75
complex : (7+5j)
```

#string

```
value="this is python programming"
print("string")
print(value)
```

Output :-

```
In [4]: runfile('C:/Rutuja patil/python1/string.py', wdir='C:/Rutuja patil/python1')
string
this is python programming
```

#list

```
l1=[10,20,30,40]
l2=["veena","riya","manasi"]
print("list")
print("value l1 =",l1)
print("value l2 =",l2)
```

Output :-

```
In [5]: runfile('C:/Rutuja patil/python1/list.py', wdir='C:/Rutuja patil/python1')
list
value l1 = [10, 20, 30, 40]
value l2 = ['veena', 'riya', 'manasi']
```

#tuple

```
t1=(10,20,"ritss")
print("tuple")
print(t1)
```

Output :-

```
In [6]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
tuple
(10, 20, 'ritss')
```

#dictionary

```
s={"name":"sai","city":"sangli","per":"86.90"}
print("dictionary")
print(s)
```

Output :-

```
In [7]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
dictionary
{'name': 'sai', 'city': 'sangli', 'per': '86.90'}
```

b)Write a program to convert data type from int to string, float to Int , str to int

#integer to string

```
n_int=20
n_str=str(n_int)
print("int to str")
print("before",type(n_int))
print("conversion of int to string=",type(n_str))
```

Output :-

```
In [8]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
int to str
before <class 'int'>
conversion of int to string= <class 'str'>
```

#float to integer

```
n_float=49.20
n_int=int(n_float)
print("float to int ")
print("before",type(n_float))
print("conversion of float to int",type(n_int))
```

Output :-

```
In [9]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
float to int
before <class 'float'>
conversion of float to int <class 'int'>
```

#string to integer

n_str=1002

n_int=int(n_str)

print("string to integer")

print("before",type(n_str))

print("cinversion of string to integer",type(n_int))

Output :-

```
In [10]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
string to integer
before <class 'int'>
cinversion of string to integer <class 'int'>
```

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LAB EXPERIMENT 2

Title :- Study of oprerator

Q1.Write a program to input two numbers and demonstrate arithmetic operation

```
#arithmatic opration
num1=int(input("enter first number"))
num2=int(input("enter second number "))
print("1.addition\n 2.substration \n 3.multiplication \n 4.division \n 5.reminder")
ch=int(input("enter your choice"))
if ch==1:
   print("addition is :",num1+num2)
elif ch==2:
    print("substraction is :",num1-num2)
elif ch==3:
    print("multiplication is :",num1*num2)
elif ch==4:
    print("division is :",num1/num2)
elif ch==5:
    print("reminder is :",num1%num2)
else:
    print("invalid choice")
```

Output :-

```
In [1]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
enter first number 4
enter second number 5
1.addition
 2.substration
 3.multiplication
 4.division
 5.reminder
enter your choice 1
addition is : 9
```

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LAB EXPERIMENT 3

Title : Study of String method

Q. Write a program to demonstrate Built-in string method (like upper(), lower(),find(),replace(),Split(),join())

```
#built in string method

line="welcome to python programming"

print("string in upper case",line.upper())

print("string in lower case",line.lower())

print("python is at position :",line.find("python"))

print("original value",line,"after replacing the value of line",line.replace("python","java"))

print("spliting values",line.split())
```

Output : -

```
In [11]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
string in upper case WELCOME TO PYTHON PROGRAMMING
string in lower case welcome to python programming
python is at position : 11
original value welcome to python programming after replacing the value of line welcome to java programming
spliting values ['welcome', 'to', 'python', 'programming']
```

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LAB EXPERIMENT 4

Title: Study of Regular Expression and Pattern Matching

a) Check pattern is available in string using in operator Extract digit from a String

i. **Using in operator**
ii. **Using find method**
iii. **Using search method**
iv. **Extract digit from string**
v. **Input email id check if it is valid or not**

#pattern matching

```
#using in oprator
import re
text="welcome to python"
pattern="python"
if pattern in text:
    print("pattern found")
else:
  print("pattern not found")
```

Output :-

```
In [14]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
pattern found
```

#using find method

```
text="welcome to python"
pattern="python"
pos=text.find(pattern)
if pos != -1:
    print(f"pattern found at index {pos}")
else:
    print("pattran not found")
```

output:-

```
In [15]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
pattern found at index 11
```

#using search method

```
text="welcome to python"
pattern=input("enter a pattern to search")
if re.search(f"^{pattern}",text):
    print("the string starts with the given pattern")
elif re.search(f"{pattern}$",text):
    print("the string ends eith the given pattern")
```

Output :-

```
In [17]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
enter a pattern to search python
the string ends eith the given pattern
```

 # extract digit from string

```
import re

line="my name is sham and roll number is 21 and lucky number is 456"

pat1=r'\d+'

nums=re.findall(pat1,line)

print("number in line",nums)
```

Output :-

```
In [18]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
number in line ['21', '456']
```

#input email id and check its valid or not

```
import re

email=input("enter email")

pat2=r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'

result=re.match(pat2,email)

if result:

   print("valid email")

else:

      print("invalid email")
```

Output :-

```
In [3]: runfile('C:/Rutuja patil/python/email.py', wdir='C:/Rutuja patil/python')
enter email : rutuja123@gmail.com
valid email
```

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LAB EXPERIMENT 5

1) Title : study of decision making

a.Check whether the year is leap year using conditional statements.

```
y= int(input("enter year"))

if y%4==0 and y%100 !=0 or y%400==0:

   print("it is leap year")

else:

   print("it is not leap year")
```

output:-

```
In [22]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
enter year 2004
it is leap year
```

b. Check whether a character is a vowel or consonant.

```
ch= input("enter any character")

if ch=='a' or ch=='e'or ch=='i' or ch=='o' or ch=='u':

   print("character is vowel")

else:

   print("character is consonent")
```

Output :-

```
In [23]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
enter any character i
character is consonent
```

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LAB EXPERIMENT 6

Title : Demonstration of loop control statement

a. Display fibonacci series up to n terms.

```
terms=int(input("enter number of terms"))
n1=0
n2=1
print(n1,"  ",n2,end="  ")
for i in range (1,terms-1):
   n3=n1+n2
   print(n3," ",end=" ")
   n1=n2
   n2=n3
```

Output :-

```
In [24]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
enter number of terms 7
0    1  1   2   3   5   8
```

b. Program using for loop with else to check if a number is prime or not.

```
num= int(input("enter number"))

count=0

for i in range(2,num//2):

    if num%i==0:

        count+=1

if count==0:

        print(num,"is prime")

else:

            print(num,"is not prime")
```

Output :-

```
In [25]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
enter number  10
10 is not prime
```

b) Display pattern.

```
for i in range(1,5):

    print("*"*i)
```

Ouput:-

```
In [26]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
*
**
***
****
```

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LAB EXPERIMENT 7

Title : Demonstration of list Data type

i. **Create a list of 10 numbers.**
ii. **Find the maximum and minimum number.**
iii. **Find the sum and average of number in a list.**
iv. **Perform slicing and indexing operations on a list.**
  (a) **Print 8th index position no.**
  (b) **Print 3rd index position to 5th index position no.**
  (c) **Print from 6th index position no.**

v.Store add and even number in separate add even list and countit and display

```
l1=[10,20,30,40,50,60,70,80,90,22]
even=[]
odd=[]
sum=0
for i in l1:
   sum+=1
print("list=",l1,"\nmaximum value=",max(l1),"\nminimum value=",min(l1))
print("sum of all elements of list",sum,"\naverage is ",sum//len(l1))
print("slicing and indexing oprator")
print("value at 8th index position",l1[8])
print("value from 3rd to 5th index ",l1[8])
print("value for 6th bindex",l1[6:])
for i in l1:
   if i%2==0:
      even.append(i)
   else:
      odd.append(i)
print("even number list will be ",even)
print("odd number list will be ",odd)
```

Output :-

```
In [27]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
list= [10, 20, 30, 40, 50, 60, 70, 80, 90, 22]
maximum value= 90
minimum value= 10
sum of all elements of list 10
average is  1
slicing and indexing oprator
value at 8th index position 90
value from 3rd to 5th index  90
value for 6th bindex [70, 80, 90, 22]
even number list will be  [10, 20, 30, 40, 50, 60, 70, 80, 90, 22]
odd number list will be  []
```

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LAB EXPERIMENT 8

Title: Demonstration of Set Data type

a)Perform Set operation- Union,Intersection,Difference,Symmetric Difference for set

```
s1={1,2,3,4,5,6}

s2={5,6,7,8,9,10}

print("Union of sets is ",s1|s2)

print("Intersection of stes is ",s1&s2)

print("Difference Of sets is ",s1-s2)
```

Output :-

```
In [2]: runfile('C:/Rutuja patil/python/set.py', wdir='C:/Rutuja patil/python')
Union of sets is  {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
Intersection of stes is  {5, 6}
Difference Of sets is  {1, 2, 3, 4}
```

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LAB EXPERIMENT 9

Title : Demonstration of dictionary Data type

a)Create and initialize a dictionary with key-value pairs for student

```
student={"name":"rutuja","city":"sangli"}

print("dictionary of student is",student)

student["city"]="bangluru"

print("after updating city",student)

student["course"]="mca"

print("after adding course ",student)

del student["city"]

print("after deleting city key",student)

student.popitem()

print("after deleting last key",student)
```

Output :-

```
In [28]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
dictionary of student is {'name': 'rutuja', 'city': 'sangli'}
after updating city {'name': 'rutuja', 'city': 'bangluru'}
after adding course  {'name': 'rutuja', 'city': 'bangluru', 'course': 'mca'}
after deleting city key {'name': 'rutuja', 'course': 'mca'}
after deleting last key {'name': 'rutuja'}
```

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LAB EXPERIMENT 10

Title: Demonstration of Define and Calling function

a)Write a function to calculate factorial of a number

```
def factorial(n):
    fact=1
    for i in range(1,n+1):
        fact*=i
    return fact
num=int(input("enter a number"))
result= factorial(num)
print("factorial=",result)
```

Output :-

```
In [29]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
enter a number 10
factorial= 3628800
```

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LAB EXPERIMENT 11

Title: Demonstration Of Function Returning Multiple values

a)Write a function that return the area and circumference of circle

```
 import math
def circle(r):
    area=3.14*r*r
    peri=2*3.14*r
    return area,peri
r=float(input("enter radious of circle"))
a,p=circle(r)
print("area=",a)
print("perimeter=",p)
```

Output :-

```
In [32]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
enter radious of circle 30
area= 2826.0
perimeter= 188.4
```

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LAB EXPERIMENT 12

Title: Demonstration On Positional, Keyword and Default Arguments and required length argument

a) Write a Program to display student information using positional, keyword and default argument

```
 def student_info(name,course="mca",year=1):
    print("name",name)
    print("course",course)
    print("year",year)
    print("-------")
student_info("amit","bca",year=3)
student_info(name="priya",year=2,course="mba")
student_info("sneha")
```

Output :-

```
In [30]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
name amit
course bca
year 3
-------
name priya
course mba
year 2
-------
name sneha
course mca
year 1
-------
```

b)Write a function total_marks(*m) that accepts any number of marks and returns the total using variable_length arguments

```
def total_marks(*m):
    total=0
    for i in m:
        total+=i
    return total
print("total marks of three subjects=",total_marks(80,90,75))
print("total marks of two subjects=",total_marks(60,70))
```

Output :-

```
In [31]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
total marks of three subjects= 245
total marks of two subjects= 130
```

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LAB EXPERIMENT 13

Title: Demonstration on Pass by Value and Pass by Reference .

Q. Write a program to show difference between Pass by Value and Pass by Reference.

```
def change_number(n):
    n=n+5
    print("Inside change_number:",n)
def change_list(L):
    L[0]=L[0]+5
    print("Inside change_list:",L)
#main program
num=10
mylist=[10,20,30]
print('\n\n Pass By Value')
print("\nBefore calling change_number:",num)
change_number(num)
print("After calling change_number:",num)
print('\n\n Pass By Referance')
print("\nBefore callingt change_list:",mylist)
change_list(mylist)
print("After calling change_list:",mylist)
```

Output:

```
In [31]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')

 Pass By Value

Before calling change_number: 10
Inside change_number: 15
After calling change_number: 10


 Pass By Referance

Before callingt change_list: [10, 20, 30]
Inside change_list: [15, 20, 30]
After calling change_list: [15, 20, 30]
```

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LAB EXSERCISE 14

Title: Demonstration on Module.

Q. Create a user-defined module arith.py that contains functions for all arithmetic operations: addition, subtraction, multiplication, division, modulus and power. write a main program to import this module and perform operations on two numbers.

```
def add(a,b):
    return a+b
def sub(a,b):
    return a-b
def mul(a,b):
    return a*b
def div(a,b):
    if b==0:
      return "Division by zero not allowed"
    return a/b
def mod(a,b):
    return a%b
def power(a,b):
    return a**b
```

#Main.py

```
import arith
a=int(input("First no:"))
b=int(input("Second no:"))
print("Addition=",arith.add(a,b))
print("Substraction=",arith.sub(a,b))
print("Multiplication=",arith.mul(a,b))
print("Division=",arith.div(a,b))
print("Modules=",arith.mod(a,b))
print("Power=",arith.power(a,b))
```

Output:

```
First no:50
Second no:10
Addition= 60
Substraction= 40
Multiplication= 500
Division= 5.0
Modules= 0
Power= 97656250000000000
```

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LAB EXSERCISE 15.

Title: Demonstration Of Package

a) **Create a Package named shapes with two modules:**
b) **Circle.py- function to calculate area and circumference**
c) **Rectangle.py- function to calculate area and perimeter**
d) **Write a main program to import package and perform calculations**

#circle.py

```
import math
def area(r):
    return 3.14*r*r
def circumference(r):
    return 2 * 3.14 * r
```

#Rectangle.py

```
def area(l,b):
    return l*b
def perimeter(l,b):
    return 2 * (l + b)
```

#Main.py

```
from Shape import circle,rectangle
r=5
print("Circle Area:",round(circle.area(r),2))
print("Circle Circumference:",round(circle.circumference(r),2))
l=10
b=6
print("Rectangle Area:",rectangle.area(l,b))
print("Rectangle Perimeter:",rectangle.perimeter(l,b))
```

Output:-

```
circle area= 78.54
Circle Circumference= 31.42
Rectangle Area= 60
Rectangle Perimeter= (10, 6, 10, 6)
```

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LAB EXSERCISE 16

Title: Demonstration of exception handling.

Q. Write a python program to demonstrate exception handling using try-except-else block.

```
try:
    num=int(input("Enter a number:"))
    result=10/num
except ValueError:
    print("Error:Please enter a valid number: ")
except ZeroDivisionError:
    print("Error:Division by zero is not allowed.")
else:
    print("Division successful. Result=",result)
```

Output:-

```
Enter a number:1
Division successful. Result= 10.0
```

```
Enter a number:0
Error:Division by zero is not allowed.
```

```
Enter a number:abc
Error:Please enter a valid number:
```

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LAB EXSERCISE 17

Title: File Handling

Q. Open a file to store student information read the data and display and closing a file.
Again read file using readline() and readlines(), move file positions using seek(), display data.

```
f = open("student.txt", "w")
f.write("Name: Gayatri\n")
f.write("Class: MCA\n")
f.write("Marks: 95\n")
f.close()
f = open("student.txt", "r")
content = f.read()
print("---- File Content ----")
print(content)
print("\n---- File Attributes ----")
print("File Name:", f.name)
print("File Mode:", f.mode)
print("Is File Closed?:", f.closed)
f.close()
print("File closed:", f.closed)
f = open("student.txt", "r")
print("\n---- Using readline() ----")
print(f.readline())   # Reads first line
print(f.readline())   # Reads second line
f.seek(0)
print("\n---- Using readlines() ----")
lines = f.readlines()
for line in lines:
    print(line.strip())
f.seek(5)
print("\nText after position 5:")
print(f.read())
f.close()
```

Output :-

```
---- File Content ----
Name: Gayatri/nClass: MCA
Marks: 95


---- File Attributes ----
File Name: student.txt
File Mode: r
Is File Closed?: False
File closed: True

---- Using readline() ----
Name: Gayatri/nClass: MCA

Marks: 95
```

```
---- Using readlines() ----
Name: Gayatri/nClass: MCA
Marks: 95

Text after position 5:
 Gayatri/nClass: MCA
Marks: 95
```

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LAB EXSERCISE 18

Title: Demonstration of Class

Q. Create a class library to store book details (book_id, title, author, price). Write methods to add book details and display all details. Using an object.

```
class Library:
    def __init__(self):
        self.books = []

    def add_book(self, book_id, title, author, price):
        book = {
            'book_id': book_id,
            'title': title,
            'author': author,
            'price': price
        }
        self.books.append(book)

    def display_books(self):
        for book in self.books:
            print(f"ID: {book['book_id']}, Title: {book['title']}, Author: {book['author']}, Price: {book['price']}")

library = Library()
library.add_book(1, "Python Programming", "John Doe", 29.99)
library.add_book(2, "Data Science", "Jane Smith", 39.99)

library.display_books()
```

Ouput:-

```
ID: 1, Title: Python Programming, Author: John Doe, Price: 29.99
ID: 2, Title: Data Science, Author: Jane Smith, Price: 39.99
```

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LAB EXSERCISE 19

Q. write methods to calculate area and Circumference. Create multiple objects and display result.

```
import math
class circle:
    def __init__(self,radius):
        self.radius=radius
    def area(self):
        return 3.14*self.radius*self.radius
    def circumference(self):
        return 2*3.14*self.radius
    def display(self):
        print(f"Radius:{self.radius}")

        print(f"Area:{self.area():2f}")

        print(f"circumference:{self.circumference():2f}")

        print("_____________")

c1=circle(5)
c2=circle(7)
c3=circle(10)
#display results
c1.display()
c2.display()
c3.display()
```

Output:-

```
In [16]: runfile('C:/Users/CIMDR/rutuja/circle.py', wdir='C:/Users/CIMDR/rutuja')
Enter Radius 10
Radius is  10.0
Area Of circle  314.0
Circumference  62.800000000000004
--------------
Radius is  3.4
Area Of circle  36.2984
Circumference  21.352
--------------
Radius is  5.6
Area Of circle  98.4704
Circumference  35.168
--------------
Radius is  5
Area Of circle  78.5
Circumference  31.400000000000002
--------------
```

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LAB EXSERCISE 20

Title: Single Inheritance

Q. Create a class employee with emp_id and salary.

Create a subclass manager that add department using single inheritance. Display employee and manager information.**

```
class employee:
    def __init__(self,emp_id,salary):
        self.emp_id=emp_id
        self.salary=salary
    def show_employee(self):
        print("Employee Id:",self.emp_id)
        print("Salary:",self.salary)
class manager(employee):
    def __init__(self,emp_id,salary,department):
        super().__init__(emp_id,salary)
        self.department=department
    def show_manager(self):
        self.show_employee()
        print("Department:",self.department)
#Object Creation
m=manager(1,50000,"Testing")
m.show_manager()
```

Ouput:-

```
In [18]: runfile('C:/Users/CIMDR/rutuja/employee.py', wdir='C:/Users/CIMDR/rutuja')
enter salary  20000
Id Of Employee  101
Department  Computer
```

Title: Multiple Inheritance

Q. Create class device with brand and price.

Create class features with ram and storage derive smartphone class from device and features and display full specifications.

```
class device:
    def __init__(self,brand,price):
        self.brand=brand
        self.price=price

class feature:
    def __init__(self,ram,storage):
        self.ram=ram
        self.storage=storage

class smartphone(device,feature):
    def __init__(self,brand,price,ram,storage):
       device.__init__(self,brand,price)
       feature.__init__(self,ram,storage)
    def show(self):

print("Brand=",self.brand)
        print("Price=",self.price)
        print("ram=",self.ram)
        print("Storage=",self.storage)

s1=smartphone("vivo", 25000, 8 , 64)
s1.show()
```

Output:-

```
In [19]: runfile('C:/Users/CIMDR/rutuja/device.py', wdir='C:/Users/CIMDR/rutuja')
Brand= vivo
Price= 25000
ram= 8
Storage= 64
```

Title: Multilevel Inheritance

Q. Write a multilevel inheritance program student -->UG Student-->PG Student.

```
class Student:
   def __init__(self,name):
      self.name=name
   def show(self):
      print("Name=",self.name)
class ugstudent(Student):
   def __init__(self,name,stream):
      super().__init__(name)
      self.stream=stream
   def show_ug(self):
      self.show()
      print("Stream=",self.stream)
class pgstudent(ugstudent):
   def __init__(self,name,stream,speci):
      super().__init__(name,stream)
      self.speci=speci
   def show_pg(self):
      self.show_ug()
      print("Specification ",self.speci)
s1=pgstudent("rutuja", "Computer", "python")
s1.show_pg()
```

Output:

```
In [23]: runfile('C:/Users/CIMDR/rutuja/student.py', wdir='C:/Users/CIMDR/rutuja')
Name= rutuja
Stream= Computer
Specification  python
```

Title: Hierarchical Inheritance

Q. Write a program using hierarchical inheritance where class vehicle is the parent and car and bike are child classes. Display the details.

```
class Vehicle:
    def __init__(self,brand,price):
        self.brand=brand
        self.price=price

class Bike(Vehicle):
    def __init__(self,brand,price,fueltype):
        super().__init__(brand,price)
        self.fueltype=fueltype

    def show(self):
        print("Brand=",self.brand)
        print("Price=",self.price)
        print("Fuel=",self.fueltype)

class Car(Vehicle):
    def __init__(self,brand,price,engine):
        super().__init__(brand,price)
        self.engine=engine

    def display(self):
        print("Brand=",self.brand)
        print("Price=",self.price)
        print("Engine=",self.engine)

c1=Car("Ertika",500000,55)
b1=Bike("Hero",120000,44)
c1.display()
b1.show()
```

Output:-

```
In [22]: runfile('C:/Users/CIMDR/rutuja/cars.py', wdir='C:/Users/CIMDR/rutuja')
Brand= Ertika
Price= 500000
Engine= 55
Brand= Hero
Price= 120000
Fuel= 44
```

Name :- Rutuja uday patil
Roll no :- 35

LAB EXSERCISE 21

Title: Method Overriding

Q. Parent class employee has show_details(), Child class manager overrides it but also calls parent method.

```
class Employee:
    def __init__(self,id,salary):
        self.id=id
        self.sal=salary

    def show(self):
        print("enter salary ",self.sal)
        print("Id Of Employee ",self.id)

class Manager(Employee):
    def __init__(self,id,salary,dep):
        super().__init__(id,salary)
        self.dep=dep

    def display(self):
        self.show()
        print("Department ",self.dep)
m=Manager(101,20000,'Computer')
m.display()
```

Output :-

```
In [18]: runfile('C:/Users/CIMDR/rutuja/employee.py', wdir='C:/Users/CIMDR/rutuja')
enter salary  20000
Id Of Employee  101
Department  Computer
```

Name :- Rutuja uday patil
Roll no :- 35

LAB EXSERCISE 22

Title: Operator Overloading

Q. Create a class time with hours and minutes. Overload + to add two time objects.

```
class time:
   def __init__(self,hours,minutes):
      self.hours=hours
      self.minutes=minutes
   def __add__(self,other):#Overloading +
      x=self.hours+other.hours
      y=self.minutes+other.minutes
      if y>=60:
          x=x+1
          y=y+1
          return time(x,y)
   def display(self):
       print(f"{self.hours} Hours,{self.minutes}Minutes")
t1=time(2,45)
t2=time(3,30)
t3=t1+t2 #Using overloaded + operator
print("Time t1:")
t1. display()
print("Time t2:")
t2.display()
print("Time t3(t1+t2):")
t3.display()
```

Output:-

```
Time t1:
2 Hours,45Minutes
Time t2:
3 Hours,30Minutes
Time t3(t1+t2):
6 Hours,76Minutes
```

Name :- Rutuja uday patil
Roll no :- 35

LAB EXSERCISE 23

Title:-Multithreading
Q. Program using multithreading to print numbers from 1 to 10 in two threads.

```
import threading
import time
def print_num(thread_name):
    for i in range(1,11):
        print(thread_name,":",i)
        time.sleep(0.2)
t1=threading.Thread(target=print_num,args=("thread 1",))
t2=threading.Thread(target=print_num,args=("thread 2",))
t1.start()
t2.start()
t1.join()
t2.join()
```

Output :-

```
In [33]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
thread 1thread 2 : 1
 : 1
thread 1thread 2 : 2
 : 2
thread 1thread 2 : 3
 : 3
thread 2 : 4
thread 1 : 4
thread 1thread 2 : 5
 : 5
thread 2 : 6
thread 1 : 6
thread 1thread 2 : 7
 : 7
thread 1thread 2 : 8
 : 8
thread 2 : 9
thread 1 : 9
thread 2thread 1 : 10
 : 10
```

Name :- Rutuja uday patil
Roll no :- 35

LAB EXSERCISE 24

Title: Database Access

Q. Create a table student (Roll no, name, age, city) insert 5 records. Change age of student having rollno 1 as 20 delete records having roll no2.

```
import sqlite3
# ----------------CONTENT TO DATABASE --------------
con=sqlite3.connect("collage.db")
cur=con.cursor()
# ---------------------CREATE TABLE -----------------
cur.execute("""create table if not exists student(roll integer primary key ,name text,age integer,city
text)""")
print("table created succesfully")
cur.execute("insert into student values(101,'rutuja',21,'sangli')")
cur.execute("insert into student values(102,'raj',24,'satara')")
cur.execute("insert into student values(103,'ritika',20,'pune')")
cur.execute("insert into student values(104,'rahul',30,'nashik')")
cur.execute("insert into student values(105,'rutuja',22,'solapur')")

print("record inserted ")

cur.execute("select * from student")
rows= cur.fetchall()
for r in rows:
    print(r)

  # ---------------- UPDATE RECORD -------------------------
    cur.execute("update student set age=20 where roll=101")
con.commit()
print("\n record updated for roll=101")

# --------------------- SELECT ALTER UPDATE ----------------
print("\records after updates :")
cur.execute("select * from student")
for r in cur.fetchall():
    print(r)



 # -------------------------- DELETE RECORDS ---------------------------
```

```
cur.execute("delete from student where roll=101")
con.commit()
print("\n record will roll=101 delete")

cur.execute("select * from student")
for r in cur.fetchall():
    print(r)

# ------------------------- CLOSE CONNECTION ---------------------
con.close()
print("\ndatabase connection closed")
```

Output :-

```
In [34]: runfile('C:/Rutuja patil/python/p1.py', wdir='C:/Rutuja patil/python')
table created succesfully
record inserted
(101, 'rutuja', 21, 'sangli')
(102, 'raj', 24, 'satara')
(103, 'ritika', 20, 'pune')
(104, 'rahul', 30, 'nashik')
(105, 'rutuja', 22, 'solapur')

 record updated for roll=101
ecords after updates :
(101, 'rutuja', 20, 'sangli')
(102, 'raj', 24, 'satara')
(103, 'ritika', 20, 'pune')
(104, 'rahul', 30, 'nashik')
(105, 'rutuja', 22, 'solapur')

 record will roll=101 delete
(102, 'raj', 24, 'satara')
(103, 'ritika', 20, 'pune')
(104, 'rahul', 30, 'nashik')
(105, 'rutuja', 22, 'solapur')

database connection closed
```

Name :- Rutuja uday patil
Roll no :- 35

LAB EXSERCISE 25

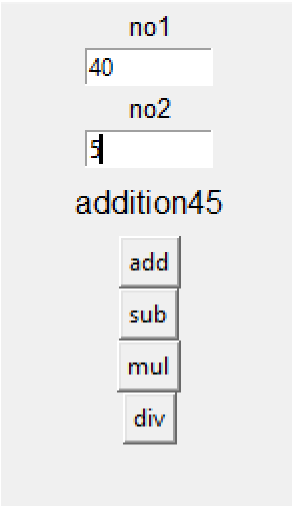
Title : GUI Program

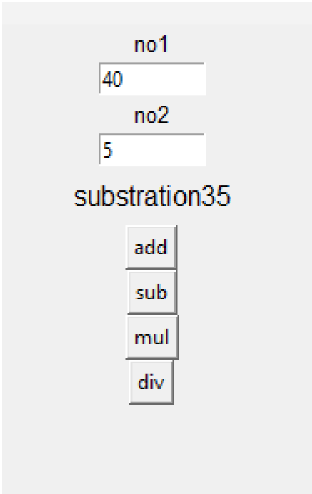
Q. GUI program for arithmetic operations.

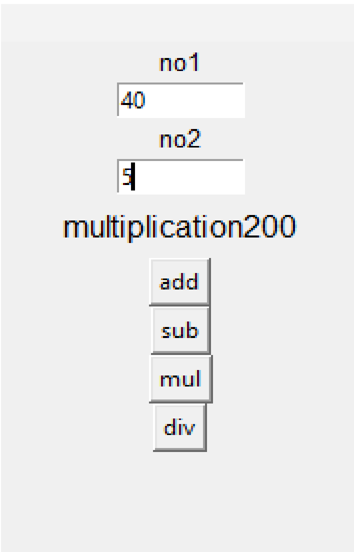
```
from tkinter import*
from tkinter import messagebox
root=Tk()
def add():
  x= int(a.get())
  y= int(b.get())
  p= x+y
  lbl_res.config(text="addition"+str(p))
def sub():
  x= int (a.get())
  y= int(b.get())
  p= x-y
  lbl_res.config(text="addition"+str(p))
def mul():
  x= int (a.get())
  y= int(b.get())
  p= x*y
  lbl_res.config(text="addition"+str(p))
def div():
  x= int (a.get())
  y= int(b.get())
  p= x/y
  lbl_res.config(text="addition"+str(p)
root.title("arithmatic operations")
Label(root,text="no1",font=("arial",10)).pack()
a=Entry (root,width=10)
a.pack()
Label(root,text="no2",font=("arial",10)).pack()
b=Entry (root,width=10)
b.pack()
lbl_res=Label(root,text="result",font=("arial",12))
lbl_res.pack(pady=5)


Button(root,text="add",command=add).pack()
```

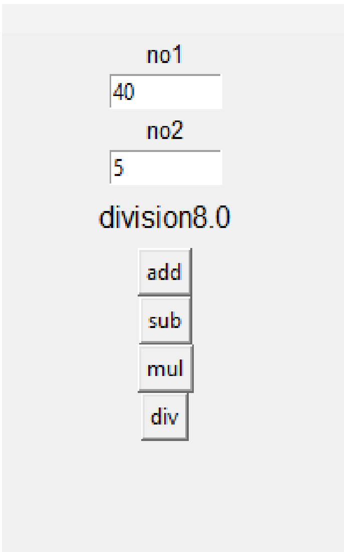
```
Button(root,text="sub",command=sub).pack()
Button(root,text="mul",command=mul).pack()
Button(root,text="div",command=div).pack()
root.mainloop()
```

Output :-









b. GUI program to input student information (roll_no, name, mobile, gender and hobbies) and display it in messagebox.

```
from tkinter import*
from tkinter import messagebox
def submit():
    name=name_var.get()
    roll=roll_var.get()
    mobile=mobile_var.get()
    gender=gender_var.get()
    hobbies=""
    if hobbies1.get()==1:
        hobbies+="Reading"
    if hobbies2.get()==1:
         hobbies+="Music"
    if hobbies3.get()==1:
         hobbies+="Sports"
    info=f"Name:{name}\nRoll
No:{roll}\nMobile:{mobile}\nGender:{gender}\nHobbies:{hobbies}"
    messagebox.showinfo("Student Details",info)
def clear():
     name_var.set("")
     roll_var.set("")
     mobile_var.set("")
     gender_var.set("")
     hobby1.set(0)
     hobby2.set(0)
     hobby3.set(0)
root=Tk()
root.title("Student Information Form")
root.geometry("400*450")
name_var=StringVar()
roll_var=StringVar()
mobile_var=StringVar()
gender_var=StringVar()
hobby1=IntVar()
hobby2=IntVar()
hobby3=IntVar()
Label(root,text="Student Information Form",font=("Arial",16,"bold")).pack(pady=10)
Label(root,text="Name").pack()
Entry(root,textvariable=name_var,widh=30).pack()
Label(root,text="Roll No").pack()
Entry(root,textvariable=roll_var,widh=30).pack()
```

```
Label(root,text="Mobile").pack()
Entry(root,textvariable=mobile_var,widh=30).pack()
Label(root,text="Gender").pack()
Radiobutton(root,text="Male",variable=gender_var,value="Male").pack()
Radiobutton(root,text="Female",variable=gender_var,value="Fmale").pack()
Label(root,text="Hobbies").pack()
Checkbutton(root,text="Reading",variable=hobby1).pack()
Checkbutton(root,text="Music",variable=hobby2).pack()
Checkbutton(root,text="Sports",variable=hobby3).pack()
Button(root,text="Submit",command=submit,width=12,bg="lightgeen").pack()
Button(root,text="Clear",command=clear,width=12,bg="lightgeen").pack()
root.mainloop()
```

Output: